

Exploratory Data Analysis (EDA) Report – Titanic Dataset

1. Introduction

This report presents an exploratory data analysis (EDA) of the Titanic dataset, which contains details of passengers aboard the Titanic. The goal is to uncover patterns and relationships among variables, particularly focusing on factors affecting survival. The dataset includes features such as age, gender, passenger class, fare, and survival status.

2. Dataset Overview

The dataset contains information on 891 passengers with 12 columns. A quick look at the data using `.info()` shows that some columns like Age, Cabin, and Embarked have missing values. Descriptive statistics reveal that:

- Age ranges from 0.42 to 80 years.
- Fare ranges from 0 to over 500.
- The average survival rate is approximately 38%.
- Most passengers traveled in the 3rd class.

3. Univariate Analysis

To understand the distribution of individual variables, histograms and boxplots were used:

- Age shows a bimodal distribution, with many young adults and middle-aged passengers.
- Fare is heavily right-skewed, indicating most passengers paid lower fares.
- Passenger Class (Pclass) and family-related features like SibSp and Parch are discrete and clustered around small values.
- Boxplots show several outliers in the Fare column, especially among higher-class passengers.

4. Bivariate Analysis

We explored relationships between features using survival as the target variable:

- Survival by Passenger Class shows that 1st class passengers had a significantly higher chance of survival than those in 3rd class.
- Survival by Gender indicates that females had a much higher survival rate compared to males.
- A scatter plot of Age vs Fare colored by Survival status shows that:
 - Children and high-paying passengers had better chances of survival.
 - There is no strong linear correlation, but some patterns are visible.

5. Correlation Matrix (Heatmap)

A heatmap was created to visualize correlations between numeric features. Key observations include:

- A negative correlation between Pclass and Survived, showing better-class passengers were more likely to survive.
- Fare has a slight positive correlation with survival.
- Very weak or no correlation was found between Age and Survived.

6. Pairplot Observations

Using Seaborn's pairplot, we visualized relationships between Pclass, Age, Fare, Sex_numeric, and survived. The pairplot shows:

- Clustering of survivors by gender and class.
- Survivors tend to be in the first class and female.
- The visual density in plots helps identify overlap and separation in variables with respect to survival.

7. Key Findings

- Gender and class strongly influence survival—females and higher-class passengers survived more.
- Young children and passengers with higher fares had better chances of survival.
- Fare shows extreme values and should be handled carefully during modeling.
- Most passengers were in 3rd class, but they had the lowest survival rate.
- Missing values exist in key features and need appropriate treatment before further modeling.